

Iheb Elleuch

B.Eng. Automation Systems and Production Informatics

Automation Engineering graduate specializing in industrial and hardware-near software development, visualization, and monitoring. Explored LLM-based solutions for writing process files in laser manufacturing with limited data.

Proven ability to structure and plan projects, work independently, and deliver results in a timely manner. Current Master Student at the University of Stuttgart in Engineering Cybernetics, focusing on Machine Learning, Robotics, and Control.

elleuchiheb2002@gmail.com +49 176 84848534 linkedin.com/in/iheb-elleuch elleuch-tech.de 73734 Esslingen

EDUCATION

Master of Science since 10/2026
University of Stuttgart (Stuttgart, Germany)
Engineering Cybernetics (1. Semester)

- **Focus Areas:** Concepts of Automatic Control (Lyapunov Stability, LQR, $\mathcal{H}_\infty$), Modeling & Simulation in Mechatronics, Machine Learning in System Dynamics, Control Lab (3DOF Helicopter).

Bachelor of Engineering 09/2022 - 02/2026
Esslingen University of Applied Sciences (Göppingen, Germany)
Automation Systems and Production Informatics

- **Final Grade:** 1.9 (210 ECTS)
- **Core Qualifications:** Real-time software development, distributed automation solutions, and industrial networking (PLCs, fieldbuses, motion control, and remote diagnostics).
- **Engineering Scope:** Hardware/software co-design, simulation, and commissioning of mechatronic systems through interdisciplinary project engineering.
- **Semester Abroad:** ESTACA, Paris, France (01/2025 - 05/2025)
- **Extracurriculars:** Co-founder of the underwater robotics initiative U.Stall

PROFESSIONAL EXPERIENCE

Bachelor Thesis in Laser Technology 09/2025 - 02/2026
Bosch Manufacturing Solutions GmbH

- Demonstrated the feasibility of LLMs to assist in the creation of process files in industrial environments.
- Overcame limited data availability through synthetic data scaling techniques and the development of a benchmark method.
- Evaluated fine-tuning techniques and data curation methods to optimize performance on local hardware.

Research & Software Engineering Intern 09/2024 - 01/2025
Bosch Manufacturing Solutions GmbH

- Led the UI/UX development of a C#/WPF graphical interface for dynamic laser beam shaping.
- Designed a robust backend communication architecture using gRPC and Python.
- Expanded backend functionality to support new features and optimized laser beam shaping algorithms.
- Conducted routine validation of the software prototype.

Working Student - Real-Time Systems 12/2022 - 08/2024
Schuler Pressen GmbH (Andritz)

- Expanded the monitoring infrastructure and visualized critical data through Grafana dashboards.
- Collected and optimized data pipelines using Prometheus, PostgreSQL, and Node-RED routing.
- Developed and maintained time-saving automation scripts for faster data processing and easier upgrade iterations.

CORE COMPETENCIES

- **Programming:** C++, C#, Python, JavaScript, SQL, MATLAB/Simulink, VBA
- **Technologies:** WPF, .NET, gRPC, Linux, Node-RED, Grafana, Prometheus, PostgreSQL, LLM Fine-tuning
- **Robotics & Control:** Hardware-in-the-Loop, Sensor/Actuator Integration, PLCs, Motion Control, Real-Time Systems
- **Tools:** Git (GitHub, GitLab, Azure), Docker, Visual Studio, VS Code, Unity Engine, Axolotl, Llama-Factory
- **Other:** Driver's License (Class B)

LANGUAGES

- **German:** Native
- **Arabic:** Native
- **English:** C1 Professional
- **French:** B2 Intermediate

KEY PROJECTS

- **U.Stall (MATE ROV):** Software Lead & Team Lead (One Season). Architected and developed the complete software stack for vertical integration between the ROV and the pilot interface. International participation: USA 2024 (Pioneer), USA 2025 (Explorer), Canada 2026 (Explorer).
- **Academic Robotics Projects:**
  PinguBlocks: Co-developed a voice-controlled, penguin-shaped robot utilizing a Raspberry Pi and a custom web interface.
  KlemmbausteinIoT: Co-engineered a modular IoT robotics platform using interlocking brick components and commercial hardware for academic teaching environments.